

Comparative Analysis of Diffusion Models and CycleGAN for Robust Data Augmentation in Real-Time Plant Disease Detection.

Abstract. The applications of the AI implementation and computer vision are one of the latest trends that have arisen in previous years. It has a vast potential to benefit the agricultural sector. Therefore, this review noted that there are still many operational challenges. More recent attentions have focused on real data of the field. Encountered problems include field data that is often unbalanced and poorly labeled. In order to obtain wider conceptions of how these problems are being covered, This paper introduced a systematic literature review of the generative adversarial networks (GANs) and diffusion models. Identified through the literature published 37 papers out of 500 papers published between 2020 - 2026 in big databases. It has been conducted according to the PRISMA standards. The indication of such a shift, it is evident that more context-specific techniques of data augmentation are needed. In these findings, the models of CycleGAN were chosen as the best one due to their speed and mobile system compatibility. Unlike the Diffusion Models which were more lifelike to the visual dimension, especially in the case of few-shot tasks. However, This also found that diffusion-based methods are disadvantaged by their computational complexity. Optimisation Hybrid generative models are to be made computationally efficient like GANs. The latent-space diffusion systems should be optimized to support edge and mobile deployment. Extensive connection to IoT-controllable robotic systems must allow real-time data augmentation to be adaptable in mobile farming settings. Other suggestions entail the use of explainable AI (XAI). This guarantees pathological applicability of synthetic samples. It is also important to improve disease-severity grading by regression-conditional generative models. The measurement metrics of agricultural precision, image quality and real time performance should be measured as a unit. Future research needs to be done in big field validation of various crops. They are also supposed to be aware of diverse climatic conditions. This contributes towards domain shift reduction. It enhances the reliability of the plant disease detection systems powered by AI. Altogether, This paper has argued that the state of the existing methods, as well as the direction of future development to make AI-based plant disease detection more claimable in the field.

Keywords: Data augmentation · deep learning · diffusion models · generative adversarial networks (GANs) · plant disease real-time detection · agriculture

1 Introduction

Timely and effective planting in the quest to achieve sustainable agricultural production. There is a relation between food security, economic stability, and disease detection of various plants. Diseases cause up to 20-40% overall loss of crop per year and the resultant driving of the use of pesticides towards unsustainability [1, 2]. The Convolutional neural networks and deep learning technologies are used in the field of disease detection such as with VGG16, ResNet, and MobileNet [3]. The PlantNet datasets especially in the benchmark of PlantVillage have recorded over 98% accuracy, but in-field implementation is slow [4, 5]. Low quality of quality labelled images is the main problem [6] and it does not only leads to overfitting of the model although it also leaves it incapable of working correctly in real-life scenarios (the few-shot problem) [2, 7].

Common methods of augmentation techniques such as flipping, cropping, rotating, colour transformations and blurring could only increase the amount of dataset. It does not offer much diverse and realistic data [5, 8] whereas CycleGAN as a GAN-based data augmentation which comes under deep learning approaches are a widely used method for the improvement of classifiers by following the synthesis of diseased-leaved images from the healthy ones, enabling higher classifier durability without paired training data [4, 9]. Conversely, utilization of generative adversarial networks (GANs) often exhibits discrepancies with modes collapsing and volatile training, and their synthesized variability is quite limited [2, 4]. Diffusion Models (DMs), on the other hand, have become a more reliable option, as they generate the images of high quality by reversing the process of noise diffusion and they have also demonstrated better diversity and robustness in tasks related to vision compared to GAN [7, 10]. Moreover, the detector which is employed here is the YOLO (You Only Look Once) variant with improved detection speed and accuracy and allows on-device, low-latency diagnosis despite the fact that their detection accuracy is proportional to the richness and varied training data under complex field conditions [11, 12]. This study, therefore, contrasts Diffusion Models and CycleGAN on the augmented data in real-time plant disease detection in terms of accuracy and resistance to the varying aspects of the environment in the actual agricultural conditions, thus enhancing them. As illustrated in Figure 1,

1.1 Background and Related Works

The approaches of deep learning have transformed the field of plant disease diagnosis and utilization of generative data augmentation methods has become a fundamental factor in the process. This strategy makes up for the lack of data in the identification. Automated leaf inspection facilitates prompt treatment and by incorporating acquisition, preprocessing, and classification, it promotes sustainable agriculture [14, 15]. PlantVillage has enabled optimized results in disease diagnosis, however it has limitations as artificially controlled conditions are not capable of simulating in varying environmental light and obstructions encountered in the real world, creating domain gap [4]. As a result, contemporary

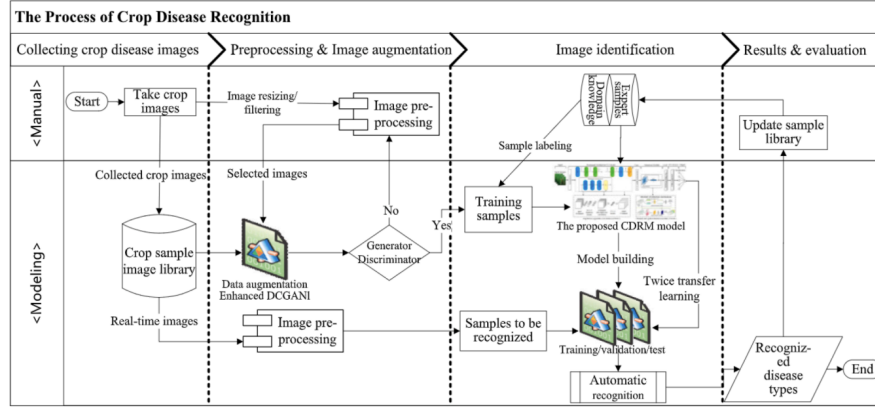


Fig. 1. Workflow of plant disease detection using deep learning for real-time diagnosis. Adapted from Chen et al. [13].

systems have been optimized from using manual features to fully automated convolutional neural networks (CNNs) [2, 5, 15].

Through the use of generative methods, overfitting can be decreased with inadequate datasets of crops collected from different areas [16]. The classical transformation of augmentation methods is to rotate and scale, these add a missing baseline but do not account for actual symptom development or compound scenery [11]. Data-driven generative augmentation leads to the synthesis of shapes that biologically plausible images imitate in the same way as the target data distributions. Both generative adversarial networks (GANs and diffusion models) capture high-level semantic features and textures, rather than relying on simple pixel perturbations, offering a more advanced approach to modeling complex agricultural environments [2, 10].

Diffusion models are important for stable training and producing images of high quality through the reversal of a slow noising process. Regulatory models like regression-conditional allow contraindications to disease severity while latent conditioning increases overall diversity [10, 17, 18]. Even though sampling is computationally expensive in the case of having multiple channels instead of ‘single-pass’ implementations of GANs, the resulting synthetic data with precise margin is preferred in this particular cultivation technique. Likelihood-based models are a strong option for generating high-dimensional, realistic datasets when compared to traditional adversarial frameworks.

CycleGAN is mainly used to unmap the infection shader cycles and adversarial losses via the cycle-consistency principle which makes it a standard method for the healthy-to-diseased leaf mapping [2, 4, 9]. The approach emphasizes the preservation of geometric structures, but some problems such as mode collapse and defects still exist. In a similar fashion, convolutional neural networks (CNNs) and transfer learning with architectures such as ResNet or MobileNet obtain a

high degree of accuracy, but often face difficulties due to the heterogeneous nature of the field data [4, 5, 15]. The latest trends suggest that the integration of lightweight models with IoT edge-cloud systems and Vision Transformers is the way forward, especially among Internet of Things. [11, 19].

2 Objectives and Research Methodology

The fast advancements in the field of generative artificial intelligence derive the need for rethinking the strategies of data augmentation in those areas that are safety-critical and require High-Precision, like agricultural intelligence [15]. A systematic literature review (SLR) is used to synthesize the fragmented evidence on the trade-offs that are incurred between GAN-based and diffusion-based synthetic data in plant pathology. This review follows PRISMA guidelines in figure 2, in order to guarantee that it provides a solid, auditable, and reliable framework for evaluating findings across the selected literature [5, 8]. The exclusion of non-English papers and defined screening criteria may lead to the influence of results [20]. A pre-established protocol together with major literature sources serves as a tool to identify the latest developments in the field of deep learning-based plant disease detection [5, 8].

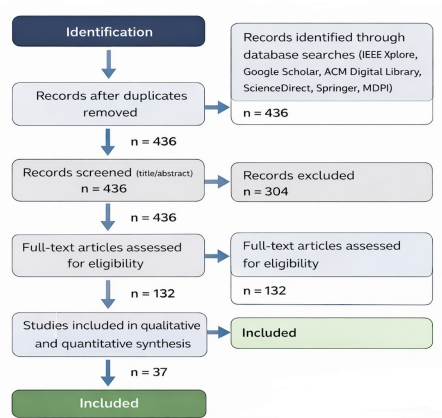


Fig. 2. PRISMA flow diagram of study selection process

2.1 Research Goals and Research Questions

The aim of this study is to address the existing between the generation of high-fidelity synthetic images and their actual implementation in autonomous agricultural systems [10]. While the performance in complex field conditions beyond PlantVillage remains an open question [4]. The conceptual framework

Table 1. Research Questions and Methodological Focus

ID	Research Question	Primary Focus / Metric
RQ1	How do Diffusion Models compare to CycleGAN in terms of image fidelity and diversity?	FID, IS, SSIM
RQ2	What architectural modifications (latent diffusion, mask conditioning) are essential for preserving botanical structures?	Structural Integrity
RQ3	What is the effect of synthetic data on the mAP and F1-score of real-time detectors (YOLOv8, YOLOv10, SSD)?	Detection Accuracy
RQ4	What are the computational trade-offs, including training stability, inference latency, and GPU resource costs?	Edge Deployment
RQ5	How effective are generative models in simulating real-field noise and environmental complexity?	Robustness
RQ6	To what extent can these models address class imbalance and rare disease scenarios?	Generalization

proposed is aimed at explaining the three-dimensional relationship between the data generation, the performance of detection, and the domain adaptation. It includes the investigation of real-time feedback analysis of the detection accuracy and robustness with respect to the different field conditions. [8,15] These points are explored in a systematic way in terms of research questions RQ1-RQ6 as outlined in Table 1 .

2.2 Primary Study Selection

Searches were carried out in the databases like IEEE Xplore, Google Scholar Labs, ScienceDirect, ACM Digital Library, Springer, Nature, and MDPI for studies published from 2020 to 2026; these were also complemented by forward snowballing which is a technique based on reviewing high-quality reviews [20]. Using the keyword strategy over titles, abstracts, and combining the keywords with “leaf,” “plant disease,” “agriculture field,” “generative AI,” “data augmentation,” “diffusion model,” “CycleGAN,” and “GNN,” the research papers found around 500 studies at first. Then a total of 37 primary studies were finally selected. These selected studies constitute the complete reference set analyzed in this review. To ensure that it has included both the basic and the most recent generative paradigms in agricultural intelligence.

Quality Assessment and Eligibility Overview The use of English deep-learning studies on diffusion and CycleGAN has greatly facilitated the ensuring of high methodological quality. Quality assurance was done following the

Kitchenham-based protocols where a threshold of 3.0 was set, and the work of low-rigor [20] was excluded. Structured extraction of metadata, techniques, and metrics that is able to use FID or F1-score made it possible to a mixed quantitative–qualitative synthesis corresponding to research questions [5, 8, 20]. Article inclusions were written in English and talked about diffusion models, CycleGAN, or generative AI for plant disease detection data augmentation. The removal of excluded papers was based on the low quality hypothesis or the utilization of traditional statistical methods by the studies involved without deep learning models [5]. Research quality was determined as per Kitchenham et al. (2015) with eight items encompassed by a 3-point Likert scale (for example, reasoning, reproducibility, dataset quality, metrics, practicality, robustness); studies that scored under 3.0 were thrown away. Access to the most important subscribed research papers required the use of external resources [20].

2.3 Data Extraction and Synthesis

Structured forms like Scribbr are used to systematically extract data on meta-data, objectives, generative methods (CNN, GAN, Diffusion), findings, limitations, and performance metrics (FID, mAP, F1-score) in order to maintain traceability and map insights to the machine learning lifecycle and specific research questions [8, 20]. Incorporating a mixed-methods strategy is the means by which new data is synthesized and reported. The quantitative synthesis (frequency, percentages, metric comparison) for RQ1–RQ2, which are mainly focusing on the trends of SSIM and FID, and the qualitative thematic synthesis for RQ3–RQ6 for the analyze deployment challenges, robustness, and generalization in the settings of the real time agriculture is the base of the methodologies.

3 Literature Review

Research on this subject grew rapidly between 2020–2026 period. Those are showing a clear shift in agricultural computer vision. Many studies began to use Generative AI to overcome the biggest problem of Data Scarcity [8, 15]. At present, based on the recent works the majority of works are focused on the combination of adversarial frameworks, diffusion architectures, and the feasibility of running them in real-time through lightweight neural networks [10, 11]

3.1 General

Recent studies on Plant Disease Detection shows that traditional data augmentation methods are such simple geometric transformations with semantic synthesis replaced by Generative AI . [5, 8]. Although, conventional approaches have shown poor performance in practical field environments [4]. CycleGAN and its derivatives, such as LeafGAN and Insta-GAN, are used for unsupervised image-to-image translation to create diseased samples from healthy ones [4, 9]. Diffusion Models, including Stable Diffusion and Latent Diffusion Models (LDM), utilize

iterative denoising to produce high-fidelity synthetic images. [17, 18] Graded augmentation via regression-conditional diffusion has also been reported as a method to simulate specific disease severity levels [17].

Since the methods and procedures employed in the Plant Disease Detection relied on deep learning, the detection and classification architectures mostly rely on CNNs and ViTs [15]. However, ResNet and InceptionV3 may be used as general feature extractors [2, 5], whereas YOLOv11 and MobileNetV3 may offer low-latency operations on real-time operations [11, 12]. Hybrid CNN-LSTM networks can preserve temporal information [5, 15], and attention mechanisms are becoming more and more able to distinguish pathological lesions against background noise [4, 12].

With the results of the investigation of the effectiveness of image augmentation in the Plant Disease Detection, generative augmentation may have a considerable positive effect on the crucial classification and generalization results. Moreover, CycleGAN could have increased the rate of detecting the disease in pears by 7 points [21], and the attention-guided approach of LeafGAN could have increased the percentage by 7.4 points by keeping the background [4]. Therefore, the best models using diffusion trained to 0.90 F1-score on strawberry datasets [10]. Moreover, adversarial hybrid training can have raised the aerial surveillance speed by 51.1% [22], and synthetic data can construct strong decision boundaries in a small-sample environment [2, 7].

In spite of the issues with real-time detection and data augmentation, real-time plant disease detection may represent the fact that such a vital undertaking trades off the complexity of generative models with the edge latency. Still, these lightweight structures as YOLOv8 can be fast, whereas frameworks such as FCL-RTDETR and latent manifold diffusion [10] can maximize efficiency. In addition, these innovations, coupled with robotic integration of the IoT [1, 14], Such integration supports, highly accurate diagnostics (97.13) even in case of dynamic variability of the field [1, 11, 23]. As shown in Table 2

3.2 The challenges and potential solutions of applying/using GAN

Although GANs have shown great progress in agricultural data augmentation, applying them in real-world settings comes with several technical and environmental challenges. Researchers are currently exploring hybrid model designs to make these systems more convenient and stable in use. [4, 15].

Lack of Labeled Data One of the biggest issues in plant disease detection is the shortage of labeled images, especially for rare or region-specific diseases. In such cases, GANs assist in creating faked images of sick leaves using small portions of healthy ones [2, 7]. This reduces the need for time-consuming manual data collection and helps bypass seasonal constraints, particularly for crops like rice and tomatoes [24, 25].

Class Imbalance Another issue is class imbalance, where the model becomes fitting toward more common diseases. GANs can help balance datasets by increasing the number of rare class samples, such as rice leaf smut or pear diseases [12, 21]. This helps improve recall and allows better identification of both common and less frequent conditions [2, 10].

Improving GAN Stability. The primary limitation regarding GAN effectiveness is “mode collapse. The generator produces a limited variety of images, and the risk of generating artifacts that are not pathologically accurate. To address this, newer models like InstaGAN and UAE-GAN include instance-level context and uncertainty awareness, leading to more meaningful and biologically accurate outputs [4, 8, 24, 26].

Field-Level Limitations Models still struggle to perform in controlled laboratory settings and the unpredictable conditions found in real farms [4, 27]. Also GANs often struggle with issues like poor lighting, overlapping leaves, or complex backgrounds. To address this, current research is moving toward IoT-integrated robotic systems that perform real-time data collection and diagnosis. [1]. The integration of Explainable AI (XAI) such as EigenCAM and Grad-CAM is also proposed as a solution to ensure that models focus on actual leaf symptoms rather than the background clutter synthesized during the augmentation process [12, 27].

Table 2. Comparison of CycleGAN and Diffusion Models for Plant Disease Detection

Aspect	CycleGAN Strengths	Diffusion Model Strengths
Training Stability	Learns mappings between healthy and diseased domains without paired data [4, 9]. mitigates mode collapse using stabilized adversarial training [2, 8].	Stable training via reverse diffusion process, inherently avoiding adversarial instability and mode collapse [10, 17].
Generation Quality	Effective texture and color translation in regions of interest, improving classification accuracy on leaf datasets [4, 9].	Superior preservation of disease-specific features [17] and higher diversity, achieving better FID/KID scores [10, 28].
Real-Time Fit	Fast single-step image generation suitable for mobile deployment and YOLO-based real-time detectors [8, 11].	Latent and stable diffusion reduce cost, but inference remains slower than GAN-based methods [17, 22].
PDD Challenges	Limited semantic preservation of complex leaf geometry [2, 4] and structural transformations.	Higher inference latency requiring compression or latent-space optimization for edge deployment [17, 19].

4 Comparative Analysis

Trends in Agricultural Generative AI Research. Research on agricultural generative AI is showing a trend toward moving away from simple methods like convolutional neural networks (CNNs) to more complex synthetic methods [29]. The early research from 2022 focused on classifying leaves using CycleGAN. Research incorporating CNN-LSTM hybrid models came about in 2023 [30]. The years 2024 to 2025 have been noted to have models at the peak of maturity, the Elucidated Diffusion Model [31]. These models also demonstrated novel real-time integration with IoT systems. Research for 2026 and beyond is anticipated to focus on more pragmatic areas where realistic field-tested models are required. In this regard the focus will be on regression-conditional diffusion techniques and in-backbone latent feature modeling. These approaches will also help in increasing the relevance to industry.

Advancement of Data Augmentation Strategies. The practice of augmenting data has evolved from simple pixel manipulations such as flipping and rotating images to more sophisticated methods, including the semantically aware generation of synthetics [32]. While most traditional augmentation methods tend to overlook class imbalance and the scarcity of data, this has led to the use of alternatives based on GANs and Diffusion methods [33]. CycleGANs, for example, function as GANs for unpaired image translation, while more recent Diffusion models, such as DDPM and LDM, are able to achieve higher realism through stepwise denoising [34]. These processes can now create some of the more complicated scenarios found in agriculture, such as models for previously unachievable simulated diseased leaves.

Architectural Roles in Agricultural AI. Generative and detection models serve specific roles in agricultural AI workflows [35, 36]. For example, while Stable Diffusion [10] generates agricultural environment images based on text (e.g., weeds in strawberry beds), UAE-GAN [24] helps manage uncertainty in plant growth abnormalities. On the detection side, models optimized for real-time edge device performance, such as YOLOv11 and MobileNetV3, are constrained by the functional split. This provides high-quality data generation (Augmentors), in contrast to low-latency, real-time field operation (Detectors), which improves practicality in precision farming applications.

Comparative Considerations Between GANs and Diffusion Models. In the agricultural industry, manual feature engineering is being replaced with CNN-based transfer learning approaches [5, 15], while augmentation practices are evolving from classical methods to those involving GANs [4, 8]. Diffusion models do offer better photo-realistic textures and generation time [7, 10, 17], direct, in-depth comparisons with GANs remain limited [8, 15]. While GANs remain useful for fast inference in mobile settings, Diffusion Models excel in photorealistic texture synthesis but come at the expense of slower generation times. [4, 7, 10]. Recently, In-Backbone Latent Diffusion Models (LDMs) have been introduced

to potentially merge the fast execution of GANs with the accuracy of Diffusion Models [10].

5 Research Gaps and Future Research Directions

Plant disease is one of the major challenges for the generative methods. The classifier that is generated and biological implausibility are the problems of detection. The mistakes that are present in the CycleGAN implementations at the same time, a significant generalization gap is showing. [2, 4]. On the other hand, a large generalization gap exists, which is the models trained on controlled datasets such as PlantVillage have low performance in the realistic environments. [4, 15]. Also diffusion models have an advantage in image quality but because of the high computational cost, they cannot be implemented on edge devices while CycleGAN is efficient but less diverse [7, 10]. Furthermore, the field suffers from the lack of standardized metrics that combine agricultural accuracy and image quality. [2, 8]. Research on in-backbone latent diffusion is gaining momentum with a focus on augmenting it to the latent space to achieve a 6.5 times reduction in latency without compromising robustness [10]. The integration of XAI (Eigen-CAM/Grad-CAM) makes sure that synthetic data is kept pathologically grounded instead of being noise-driven [4, 12]. Furthermore, accurate disease-severity grading is now achievable with the help of regression-conditional models, going beyond binary detection. [17]. In reality, autonomous robot systems with IoT sensors can adapt data "on-the-fly" by responding to local microclimates, real time. [1, 37].

6 Conclusion

The results of this study indicate that CycleGAN remains the most suitable option for when it comes to image augmentation on real time and mobile-friendly needs, at least partially due to its image-translation capabilities without paired data [9]. On the other hand, diffusion models have shown better results in retaining visual details and handling datasets with class imbalance, [10]. Another model, LeafGAN, which uses attention mechanism, seems to improve precision, but it still struggles with unstable adversarial training dynamics [4]. Diffusion models are effective in few-shot learning settings. However, their practical use in real-time systems generally requires acceleration of the sampling process [10, 18]. Overall, from what observed here, blending the strengths of GANs and diffusion models may help balance efficiency and accuracy in plant disease detection and improved food security, but further testing in real agricultural conditions is necessary to confirm this direction [4].

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